

Causality - SoSe 2025

Preliminaries

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Last Time

Last time, we sketched the larger areas that we will discuss in the course (**causal inference** and **causal discovery**) and particularly focused on:

- Correlation vs. Causality (Reichenbach's common cause principle)
- Pearl's causal hierarchy
- Observational data vs. randomized controlled trials
- Simpson's paradox

In today's lecture, we recap some concepts from probability theory to bring everyone up to speed. In particular, we discuss:

- Random variables
- Probability density functions
- Variable transformations
- Cumulative distribution functions
- Multivariate random variables (and some properties)
- Expectation of a RV, and statistical independence

Random Variables

Random Variables

Real sources of randomness:

- X = “number of dots after throwing a dice”
- Y = “number of Higgs bosons after running the LHC”

Artificial sources of randomness: random number generators

- Pseudo-random number generators, e.g. Python's `rand` function

Definition (Random Variable)

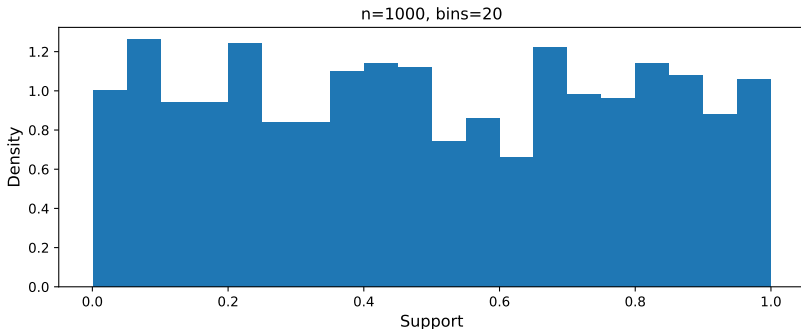
- A variable X that gets assigned the result of a *source of randomness* is called a **random variable** or **RV**.
- For concreteness, most random variables (we consider) are real-valued (including integer-valued or boolean-valued).
- The **range** $\mathcal{X}$ of a RV X is the set of possible values.

Python's random Function

With a few lines of code, we can generate pseudo random numbers drawn from a uniform distribution.

```
import numpy as np
import matplotlib.pyplot as plt

samples = np.random.uniform(size=1000)
```

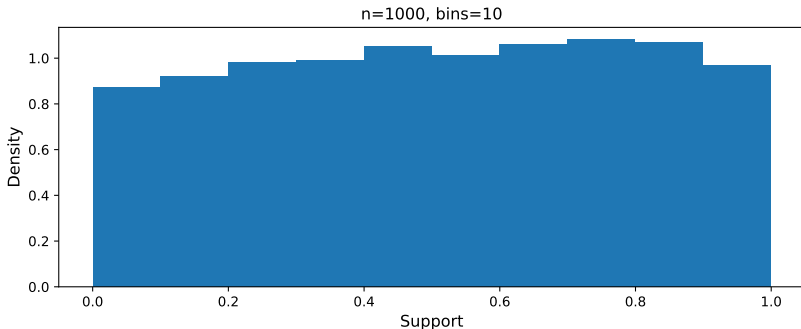


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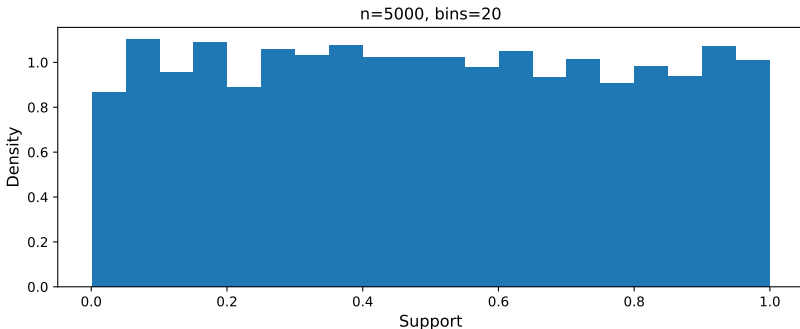


Python's random Function

With a few lines of code, we can generate pseudo random numbers drawn from a uniform distribution.

```
import numpy as np
import matplotlib.pyplot as plt

samples = np.random.uniform(size=5000)
```



Probabilities

PDF of a RV and it's Properties

- Intuitively, the **probability density function** (PDF) describes the histogram of a random variable.
- In general, we denote that RV X has PDF p_X by writing

$$X \sim p_X(x)$$

which is read: X is **distributed according to** p_X . We use small letters x for **values** of the RV X . Sometimes the PDF has a different name, e.g. u or f or simply p . The range of a real-valued RV is $\mathbb{R}$.

- A PDF $f : \mathcal{X} \rightarrow \mathbb{R}$ is **non-negative** $f(x) \geq 0$ for all $x \in \mathcal{X}$.
- The **support** is the subset of the range of X where f is non-zero,

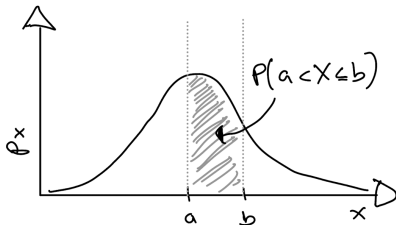
$$\text{supp}(f) = \{x : f(x) > 0\} \subset \mathcal{X}$$

- A PDF is **normalized** to integrate to one over the whole range of X ,

$$\int_{\mathcal{X}} f(x) dx = 1$$

Definition: Consider a real-valued RV $X \sim p_X$. The **probability** that the value of X is inside a certain **open interval** $(a, b]$ is the integral of the PDF p_X over that interval:

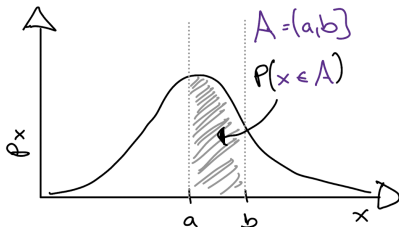
$$P(a < X \leq b) = \int_a^b p_X(x) dx$$



Definition: Consider a real-valued RV $X \sim p_X$. The **probability** that the value of X is inside a **subset** $A \subset \mathcal{X}$ of its range, is the integral of the PDF p_X over that set:

$$P(X \in A) = \int_A p_X(x) dx$$

Note that this integral is not for all subsets of $\mathcal{X}$ defined, i.e., it is only defined for **measurable** subsets.



Transformations

Transformations of Random Variables

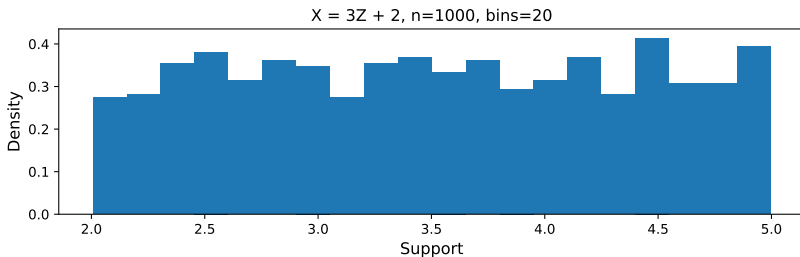
Linear transformations:

$$t_{\alpha,\beta}(z) = \alpha z + \beta$$

Consider a standard uniform random variable $Z \sim u(z)$, where

$$u(z) = [0 < z \leq 1] := \begin{cases} 1 & 0 < z \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let's look at the histogram of $X = t_{3,2}(Z)$:



Linear transformations (continued)

We can see that the **density** for each bin is **decreased** and the **support** changed, but why?

For $\alpha > 0$ one can show that the curve can be written as:

$$\begin{aligned} f_{\alpha\beta}(x) &= [0 < (x - \beta)/\alpha \leq 1]/\alpha = [\beta < x \leq \alpha + \beta]/\alpha \\ &= \frac{1}{\alpha} u\left(\frac{x - \beta}{\alpha}\right) \end{aligned}$$

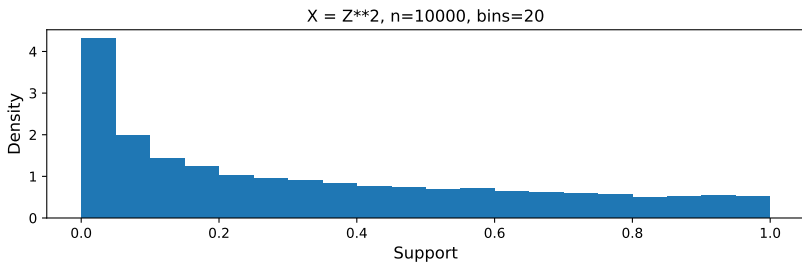
For $\alpha > 1$, the curve is below 1, for $0 < \alpha < 1$ above, **ensuring normalization**:

$$\int_0^1 f_{\alpha\beta}(x) dx = 1 .$$

Non-Linear Transformations

Now, let $t(z) = z^2$. How does it transform $Z \sim u$?

Let's look at the histogram of $X = t(Z)$:



One can show that the density $g(x)$ can be written as

$$g(x) = \frac{1}{2\sqrt{x}}[0 < \sqrt{x} \leq 1] = \frac{1}{2\sqrt{x}}[0 < x \leq 1]$$

Let's derive the general formula!

Desideratum

Probabilities should stay the same under **monotonic transformations**,
i.e. for monotonic non-decreasing t

$$P(a < X \leq b) = P(t(a) < t(X) \leq t(b)) = P(t(a) < Z \leq t(b))$$

i.e.

$$\int_a^b p_X(x) dx = \int_{t(a)}^{t(b)} p_Z(z) dz$$

The rest we will solve on the blackboard.

How does a transformation change the PDF?

Theorem Let RV $X \sim p_X$ and RV $Z = t(X) \sim p_Z$ being defined via a function $t : \mathcal{X} \rightarrow \mathcal{Z}$. Then the PDFs are related as follows:

$$p_X(x) = p_Z(t(x)) |t'(x)| = p_Z(t(x)) \left| \frac{dz(x)}{dx} \right|$$
$$p_Z(z) = p_X(t^{-1}(z)) |(t^{-1})'(z)| = p_X(t^{-1}(z)) \left| \frac{dx(z)}{dz} \right|$$

where for the latter t must be invertible.

Note that here, we use Leibniz's notation for derivatives:

$$t'(x) = \frac{dz}{dx} \qquad (t^{-1})'(z) = \frac{dx}{dz}$$

Linear transformation:

$$X \sim u(x), \quad Y = t_{\alpha\beta}(X) = \alpha X + \beta$$

Let's apply the formula:

$$\begin{aligned} p_Y(y) &= p_X(t^{-1}(y)) |(t^{-1})'(y)| \\ &= [0 \leq (y - \beta)/\alpha \leq 1] \left| \frac{1}{\alpha} \right| \end{aligned}$$

Similarly, we can solve the **non-linear** example, where $t(X) = x^2$ now:

$$p_Y(y) = p_X(t^{-1}(y)) (t^{-1})'(y) = \frac{1}{2\sqrt{x}} [0 < x \leq 1]$$

Cumulative Distribution Function

Definition (CDF) Let $X \sim p_X$. The *cumulative distribution function* (CDF) of X is defined as

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x p_X(x') dx' .$$

Properties:

- F_X is right-continuous, i.e. $\lim_{\varepsilon \rightarrow 0, \varepsilon > 0} F_X(x + \varepsilon) = F_X(x)$
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow \infty} F_X(x) = 1$
- F_X is non-decreasing (since p_X is non-negative).
- $P(a \leq X \leq b) = F_X(b) - F_X(a)$
- $F'_X(x) = p_X(x)$, i.e. **PDF is derivative of CDF**.

A (Somehwat) Advanced Aexample

Given $X \sim p_X(x)$, and let $Z = F_X(X)$, then $\mathcal{Z} = [0, 1]$.

Note: the subindex of the CDF F_X is the name of a RV, the input of F_X is a value. However, inputting a RV for a value creates a new RV.

Question: What is the PDF of Z ?

Applying the formula:

$$p_Z(z) = p_X(F_X^{-1}(z)) (F_X^{-1})'(z) = \frac{p_X(F_X^{-1}(z))}{p_X(F_X^{-1}(z))} = 1$$

where we used $F_X' = p_X$ and the **Inverse Function Theorem**, which states:

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}$$

for invertible (differtiable) function f and $b = f(a)$.

Theorem Given $X \sim p_X$ define another random variable by transforming X with its CDF,

$$Z = F_X(X)$$

then Z is uniform, i.e. $p_Z(z) = u(z) = [0 \leq z \leq 1]$.

Transforming Uniforms

Question: What happens when we **transform a uniform RV** $Z \sim u(z)$ with the inverse CDF $t = F_X^{-1}$ (the **quantile function**) of variable X ?

Let $Y = F_X^{-1}(Z)$, where $Z \sim u(z)$.

$$\begin{aligned} p_Y(y) &= p_Z(F_X(y)) (F_X)'(y) \\ &= u(F_X(y)) (F_X)'(y) \\ &= p_X(y) \end{aligned}$$

Y now has PDF, p_X !

This property is very important, e.g., in sampling. We can now **sample** $z \sim u(z)$ from **uniform** distribution, transform it with the quantile function $Y = F_X^{-1}(Z)$ of a standard normal, and **get** $Y \sim \mathcal{N}(0, 1)$.

Multivariate Random Variables

Multivariate Random Variables

Univariate RV (SO FAR)

$$X \sim p(x) \quad x \in \mathbb{R}^1 \quad \text{scalar}$$

Multivariate RV (NEW)

$$\mathbf{X} \sim p(\mathbf{x}) \quad \mathbf{x} \in \mathbb{R}^n \quad \text{vector}$$

Bivariate RV (NEW)

$$(X, Y) \sim p(x, y) \quad x \in \mathbb{R}, y \in \mathbb{R} \quad \text{pair}$$

Note: The following definitions apply to the **multivariate** case, however, for easier understanding, we consider the **bivariate** case.

Note:

- The semantics of the function p is heavily overloaded!
- Where obvious from the **name of the variable** we omit subindices:

$$p(x) = p_X(x)$$

$$p(y) = p_Y(y)$$

$$p(x, y) = p_{XY}(x, y)$$

Joint Probabilities:

- The **joint PDF** $p(x, y)$ fully describes the **two-dimensional histogram** of the pair of real-valued RVs (X, Y) .
- For pairs the **volume** (area) under the two-dimensional PDF determines the **probabilities**, e.g. for intervals:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d p(x, y) dx dy$$

where we used a “comma” instead of “and”, i.e. $P(\cdot, \cdot) = P(\cdot \wedge \cdot)$.

Definition (Joint CDF) Given a joint PDF $p(x, y)$ we define the joint CDF:

$$P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y p(x', y') dx' dy'$$

From the joint CDF back to the joint PDF

$$P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y p(x', y') dx' dy'$$

Note:

- $\lim_{x \rightarrow -\infty} P(X \leq x, Y \leq y) = 0$ for any y
- $\lim_{y \rightarrow -\infty} P(X \leq x, Y \leq y) = 0$ for any x
- In general we have: $\int_a^b f(\mathbf{t}) d\mathbf{t} = F(\mathbf{b}) - F(\mathbf{a})$ with $\frac{\partial}{\partial \mathbf{t}} F(\mathbf{t}) = f(\mathbf{t})$
- One can show:

$$p(x, y) = \frac{\partial}{\partial x} \frac{\partial}{\partial y} P(X \leq x, Y \leq y)$$

How to create univariate PDFs from a joint PDF

Joint PDF

$$p(x, y) = \dots$$

Two ways to get univariate PDF

1. Ignore Y , i.e., we do not care for Y :
 - Marginal PDF of X
 - Sum rule
2. Fix $Y = y^*$, i.e., we consider a particular value for Y :
 - Conditional PDF of X given $Y = y^*$
 - Product rule

1. Marginal PDF and Sum Rule

Definition (Marginal PDF) Given joint PDF $p(x, y)$, define the marginal PDF as follows:

$$p(x) = \frac{\partial}{\partial x} P(X \leq x, Y \in \mathbb{R}) \quad \left(= \frac{\partial}{\partial x} P(X \leq x) \right)$$

Theorem (Sum rule) Integrating out a variable from the joint PDF gives us a marginal PDF:

$$\begin{aligned} p(x) &= \frac{\partial}{\partial x} P(X \leq x, Y \in \mathbb{R}) \\ &= \frac{\partial}{\partial x} \int_{-\infty}^x \int_{-\infty}^{\infty} p(x', y) dx' dy \\ &= \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial x} \int_{-\infty}^x p(x', y) dx' \right) dy \quad (\text{Leibniz's rule}) \\ &= \int_{-\infty}^{\infty} p(x, y) dy \end{aligned}$$

2. Conditional PDF and Product Rule

Fixing $Y = y^*$ in the joint PDF, we get a non-negative function of x :

$$x \mapsto p(x, y^*)$$

which is not normalized and thus not a proper PDF. The normalizing factor is:

$$\int_{-\infty}^{\infty} p(x, y^*) dx = p(y^*)$$

Using the sum rule we get:

Definition (Conditional PDF and product rule) For y^* with $p(y^*) > 0$, the univariate PDF of X given $Y = y^*$ is defined as:

$$p(x|y^*) = \frac{p(x, y^*)}{p(y^*)} \quad (\text{product rule})$$

Here, $p(y)$ is the marginal PDF defined analogously to $p(x)$.

Sum and Product Rules

Main rules

$$p(x) = \int p(x, y) dy$$

sum rule

$$p(x, y) = p(x) p(y|x)$$

product rule

Variations

$$p(y) = \int p(x, y) dx$$

$$p(x, y) = p(y) p(x|y)$$

$$p(x|y) = \frac{p(y|x) p(x)}{p(y)}$$

Bayes rule

$$= \frac{p(y|x) p(x)}{\int p(y|x) p(x) dx}$$

combined

Transformations

So far: univariate transformations, **next:** multivariate transformations

Transformation of multivariate random variables (1)

Setup Assume function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **invertible**.

$$\begin{aligned} \mathbf{X} &\sim p_{\mathbf{X}}(\mathbf{x}) & \mathbf{x} &\in \mathbb{R}^n \\ \mathbf{Y} &= f(\mathbf{X}) & \mathbf{y} &\in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \mathbb{R}^n \end{aligned}$$

What is the PDF of $\mathbf{Y}$?

- Let's recall our univariate formula:

$$p_{\mathbf{Y}}(y) = p_{\mathbf{X}}(f^{-1}(y)) \left| \frac{\partial \mathbf{x}}{\partial \mathbf{y}}(y) \right|$$

- However, **now** $\partial \mathbf{x} / \partial \mathbf{y}$ is the **Jacobian** of f^{-1} and $|\cdot|$ is the **determinant**.
- Note that the determinant calculates the local **volume change**, which is the key factor to keep probabilities constant.
(Analogously to univariate transformation $f(x) = \alpha x$).

Example of a Transformation¹

Setup: Given a pair (X_1, X_2) with factorizing joint PDF:

$$(X_1, X_2) \sim p(x_1)p(x_2)$$

What is $p_Y(y)$ for $Y = X_1/X_2$?

Solution Observe that $y = f(x_1, x_2) = x_1/x_2$ is not invertible, and define invertible transformation:

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = g(X_1, X_2) = \begin{bmatrix} X_1/X_2 \\ X_2 \end{bmatrix}$$

With inverse:

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = g^{-1}(Y_1, Y_2) = \begin{bmatrix} Y_1 Y_2 \\ Y_2 \end{bmatrix}$$

¹From https://en.wikipedia.org/wiki/Probability_density_function#Link_between_discrete_and_continuous_distributions

Inverse transformation

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = g^{-1}(Y_1, Y_2) = \begin{bmatrix} Y_1 Y_2 \\ Y_2 \end{bmatrix}$$

Absolute determinant of Jacobian of the inverse

$$\left| \begin{array}{cc} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{array} \right| = \left| \begin{array}{cc} y_2 & y_1 \\ 0 & 1 \end{array} \right| = |y_2|$$

Transformed PDF

$$p_{Y_1 Y_2}(y_1, y_2) = p_{X_1 X_2}(y_1 y_2, y_2) |y_2| = p_{X_1}(y_1 y_2) p_{X_2}(y_2) |y_2|$$

using the sum rule to marginalize out Y_2 :

$$p_{Y_1}(y_1) = \int p_{Y_1 Y_2}(y_1, y_2) dy_2 = \int p_{X_1}(y_1 y_2) p_{X_2}(y_2) |y_2| dy_2$$

which might be further simplified for particular choice of p_{X_1} and p_{X_2} .

Discrete Random Variables

Discrete Random Variables

Definition The randomness of a discrete RV X with finite (or countably infinite) range $\mathcal{X}$ is described by a **probability mass function (PMF)** $p_X : \mathcal{X} \rightarrow [0, 1]$ that sums up to one over all possibilities, i.e.

$$\sum_{x \in \mathcal{X}} p_X(x) = 1$$

Akin to the continuous case, the PMF has the following properties:

$$p(x) \geq 0$$

$$\sum_x p(x) = 1$$

$$p(x) = \sum_y p(x, y)$$

$$p(x, y) = p(x)p(y|x)$$

$$P(X \in A) = \sum_{x \in A} p(x)$$

Independence

Independence

Definition (Independence) Two random variables X and Y are **statistically independent**, denoted by $X \perp\!\!\!\perp Y$ if their joint PDF factorizes into the marginals, i.e.,

$$p(x, y) = p(x)p(y)$$

Independence of X and Y implies:

$$p(x) = p(x|y) \quad \text{and} \quad p(y) = p(y|x)$$

Definition (Conditional independence) Two RVs X and Y are **conditionally independent** given a third RV Z ($X \perp\!\!\!\perp Y \mid Z$), if their conditional joint PDF factorizes into conditional marginals, i.e.,

$$p(x, y|z) = p(x|z) p(y|z)$$

Expectation

Definition (Expectation) The **expectation** of an expression involving random variables is the PDF-weighted integral:

$$E[f(X)] = \int f(x) p(x) dx$$

Sometimes we omit the additional bracket: $E f(X) = E[f(X)]$

Examples

- Mean: $\mu = E[X]$
- **Variance:** $\sigma^2 = E[(X - EX)^2]$
- Probabilities are expectations over the indicator function:

$$P(X \in A) = E[1_A(X)] = E[[X \in A]]$$

This lecture mainly *builds upon and extends* the material from “*Causality*” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**.

Thank you!